

Joint adaptive transfer learning network for cross-domain fault diagnosis based on multi-layer feature fusion

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ARTICLE INFO

Article history:

Received 14 November 2020

Revised 14 September 2021

Accepted 5 November 2021

Available online 25 November 2021

Keywords:

Multi-layer feature fusion

Multilinear map

Joint adaptive transfer learning

Cross-domain intelligent fault diagnosis

ABSTRACT

Traditional intelligent fault diagnosis models are required to be trained and tested under an identical probability distribution. However, the shift in data distributions is inevitable due to changes in environmental and operational conditions, which results in diagnostic performance degradation. Currently, transfer learning has been successfully applied to learn a discriminative diagnosis model in the presence of a shift. But conventional transfer learning approaches encounter obstacles without adequately considering the feature interactivity and transferable ability at different layers. In this study, a joint adaptive transfer learning framework based on multi-layer feature fusion for reliable cross-domain diagnosis is presented to address these issues. Firstly, the multilinear map is employed to implement a novel multi-layer feature fusion. This fusion is key to realizing a substantial improvement of feature representation capability and effectively embedding joint distribution of multi-layer features. Furthermore, a novel joint adaptive transfer learning (JATL) framework is devised to facilitate reliable cross-domain adaption by making utmost use of cross-domain-invariant features with a small amount of data. Experiments with different transfer scenarios on two benchmark datasets have been conducted, and experimental results demonstrate the superiority of the proposed approach.

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1. Introduction

An accurate and generalized fault diagnosis approach is capable of distinguishing machine health conditions, which is of vital importance in Prognostic and Health Management (PHM) [1–3]. Intelligent fault diagnosis (IFD) has recently received much attention by applying machine learning to machinery fault diagnosis. It has advantages such as automatic and efficient identification of fault location and type [4,5]. However, the application of IFD to real engineering scenarios meets difficulties due to prohibitive obstacles to ensuring that all samples come from the same distribution [6].

Traditional machine learning-based approaches utilize rich mechanical expertise and mathematical knowledge to design features and combine machine learning methods to recognize health states [7,8]. However, the requirement for these approaches to design appropriate handcrafted features limits the implementation of an intelligent end-to-end diagnostic system. Currently, deep learning-based approaches have attracted increasing interest in learning hierarchical representations automatically via their

multiple layers [9,10]. Among these approaches, convolutional neural network (CNN) has been widely applied in feature learning with translation-invariant characteristics and the ability to learn spatial hierarchies of patterns [11–13]. For CNN-based models, improving the feature learning capability is of crucial consequence for accurate diagnosis. Traditionally enhancing model discriminability by stacking more layers or broadening the width at each layer is prone to overfit and considerably increase calculation burden [14]. To address this bottleneck, some multi-scale structures have aroused researchers' interest to further abstract data and improve information utilization by concatenating generated features from different layers [15,16].

Unfortunately, although deep learning-based approaches have gained impressive performance, they have poor generalization ability and longer training time when directly applied to another task under a different operating condition or environment [17]. This performance deterioration arises from a critical assumption that traditional IFD models must be trained and tested under an identical probability distribution. Generally, variation in operation conditions, adverse environmental influences, and component quality result in distribution discrepancies [16]. If a new diagnosis network is trained from scratch, sufficient labeled data containing complete knowledge about all health states is required, which is

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Nomenclature

PHM	Prognostic and health management	D	The domain of X
IFD	Intelligent fault diagnosis	$P(X)$	The marginal probability distribution of X
CNN	Convolutional neural network	χ	The feature space of X
MSCNN	Multi-scale convolutional neural network	T	A diagnosis task
MJDCNN	Multi-layer joint distribution-based CNN	Y	The label space of T
JATL	Joint adaptive transfer learning	$\phi(\cdot)$	The objective predictive function of T
GAP	Global average pooling layer	φ	The activation function
ReLU	Rectified Linear Unit	x	The instantiation of X .
1D	One-dimensional	$\mathcal{H}$	The reproducing kernel Hilbert space
CWRU	Case Western Reserve University	v^L	The convolutional outputs at the layer L
IF	Inner raceway fault	b^L	The bias vector of the filter kernels at the layer L
BF	Ball element fault	K^L	The weight of the filter kernels at the layer L
OF	Outer raceway fault	$*$	The convolution operator.
OI	The composite fault of OF and IF	l	The number of kernels
CF	Cage fault	k	A kernel in a Hilbert space
Con	The standard convolutional layer	$\mathbb{R}$	The set of real numbers
MP	The max-pooling layer	F	The feature extractor layer
TCNN	Transferable convolutional neural network	W	The signal channels of the input signal
HP	Horsepower	y	The output feature maps of F
AUC	Area under curve	c	The length of y
$\langle \rangle_H$	An inner product on $\mathcal{H}$	w	The signal channels of y
z	The infinite-dimensional feature map	M	The mapping layer
$\mu_X(P)$	The expected feature map of $P(X)$	G	The output layer
C	The length of the input signal	h	The joint variable
g	The GAP output of y	$C_h(P)$	The distribution of h
θ	The parameters in MJDCNN	t	The label of the input sample x
$\hat{\theta}$	The optimal value of θ	E	The cross-entropy loss
f_{Adam}	The computation functions in the Adam optimizer for θ	JD_{ij}	The multilinear features of the i^{th} and j^{th} layer
S	The source subset	ε	The penalty coefficients
TP	The number of true positive instances	T	The target subset
FN	The number of false-negative instances	δ	The learning rate
FP	The number of false-positive instances		
X	The random input variable		

tricky in industrial practice [18]. During most of its operating phases, the machinery runs in a normal condition, and faults rarely occur [17]. Meanwhile, the degradation process is too long to collect complete fault samples of all health states [17]. Thus, there is a strong incentive to build an effective IFD learner with strong generalization. It can utilize rich annotated data from another dataset (source domain) for a different but related testing dataset (target domain) under evolving working conditions or environments.

Nowadays, transfer learning-based approaches are promising to solve the above learning difficulties [19]. They aim to learn a discriminative diagnosis model by leveraging rich knowledge from another related task and reducing the shift in data distributions [20]. Instead of training a diagnosis model from scratch, the most popular trick in transfer learning is to reuse and fine-tune pre-trained models to a new diagnosis task with a small amount of labeled data [21]. It transfers diagnosis knowledge by discovering shared parameters or priors of hyperparameters between diagnosis models of the source and target tasks [19]. Shao et al. [22] made use of pre-trained convolutional bases in the well-known VGG-16 network and presented a cross-domain fault pattern recognition approach of bearings and gearboxes. Li et al. [23] pre-trained CNN and multi-layer perceptron and transferred them to target data under different working loads and component types. Zhao et al. [16] proposed a multi-scale convolutional neural network (MSCNN) to extract various transferable features through different-scale receptive fields. Chen et al. [21] proposed a deep transfer CNN framework with a fine-tuned fully connected layer in the target domain. Intuitively, the crucial issue in these

approaches is to preserve information and fine-tune networks appropriately. As a result, some fine-tuning schemes like full-tuning and layer-wise fine-tuning schemes are adopted to modify pre-trained parameters to new tasks adaptively [24]. Some layers are frozen in these approaches, whose parameters are fixed. Despite the above various learning-based diagnosis models that have already been explored, there are still two main limitations for the practical application that can be studied and improved:

- 1) Previous studies mainly focus on constructing a micro neural network to enhance feature learning capabilities of local patches but have strict requirements for feature shape. However, as the level of a neural network deepens, the abstraction degree of extracted representation, the activation response mechanism, and the shape of features are varying, an efficient and flexible feature fusion between layers over a longer distance is needed to avoid the loss of early statistics information [25].
- 2) The transferability of different levels in a network is rarely investigated in the area of fault diagnosis. Deep convolutional networks can provide cross-domain deep features, but layer by layer, their transferability decreases from general to specific [26]. General features are cross-domain-invariant to other tasks, while the others appear more specific to a particular task or dataset. Direct full-tuning of the whole network can destroy valuable general information obtained in the pre-trained model. Layer-wise fine-tuning schemes try to overcome this obstacle by freezing some lay-

ers from fine-tuning until higher layers are relatively adaptive to the new task. But because of the limitation of sequential network architecture and the influence of frozen intermedia layers, the general features cannot effectively assist the fine-tuning of specific levels [27].

Thus, we propose a joint adaptive transfer learning (JATL) framework to solve these limitations in cross-domain diagnosis. A multi-layer joint distribution-based CNN (MJDCNN) for flexible and effective feature fusion with better interactions between layers is exploited to overcome the first weaknesses. It captures correlations between multi-layer features and performs great feats of embedding their joint distribution using the multilinear map. To overcome the second weakness, an excellent JATL framework is devised considering varying transferable abilities at different layers. The developed framework fine-tunes specific layers combining cross-domain-invariant prior information conveyed in the general features based on the multi-layer joint distribution. This transfer learning framework effectively serves the pre-trained network as a reused model for the new task and only needs limited data to fine-tune it.

The remainder of this paper is structured as follows: Section 2 describes the focused problem and some fundamental theories about the proposed approach. The MJDCNN and proposed transfer learning framework are described in Section 3. Section 4 presents details of the used datasets and an experimental evaluation of the proposed approach. A comparison of results is presented in section 5, along with some discussions. Section 6 draws some conclusions and future work.

2. Problem description and theoretical background

In this section, feature characteristics, the transfer learning problem, and theories of the proposed neural network architecture and Hilbert space embedding are provided in detail.

2.1. Feature characteristics and the transfer learning problem

Deep networks establish the mapping from a high-dimensional, rich signal to health states. Impressively, deep neural networks exhibit a phenomenon: they are prone to learn features containing statistical information in signals at shallow layers [25]. Meanwhile, increasingly abstract features are learned and mapped to larger regions in the original input as the hierarchy goes deeper. Lately, the research about simultaneously enabling the diagnosis model to acquire deeper abstract and early statistical information has become a hotspot [28].

Moreover, with the above feature characteristics in deep networks, some pursuits of optimal transfer learning are conducted. In the transfer learning field, a domain D consists of a marginal probability distribution $P(X)$ and feature space χ , where the random input variable $X \in \chi$. Each task T consists of a label space Y and an objective predictive function $\phi(\cdot)$ for predicting the corresponding health state [19]. The shift in distributions occurs when the source data X_S and target data X_T are drawn from different marginal probability distributions $P(X_S)$ and $P(X_T)$. Furthermore, recent evidence suggests that as the layer goes deeper, the transferability of convolutional features extracted from different layers will transit from general to specific [27]. General features capably represent other data sets, and specific features are related to a particular data or task where the distance between the source and target tasks also affects the transferable ability at different layers [26]. The analysis of feature characteristics and transferability helps apply IFD to real scenarios. In practical engineering occasions, two obstacles to applying traditional IFD approaches can be sum-

marized: 1) it is costly to collect sufficient labeled data containing all health status information to retrain a new learning model, 2) because of the discrepancy between domains, directly reusing deep networks to new datasets or environments usually have weak performance. Specifically, to address deficiencies in cross-domain diagnosis, transfer learning is employed to improve the learning performance of the predictive function $\phi_T(\cdot)$ of a targeted task T_T by transferring knowledge from the source task T_S in this paper. However, the variation in transferability imposes challenges to perform reliable transfer processes in cross-domain diagnosis. Therefore, a novel joint adaptive transfer learning framework based on multi-layer feature fusion is proposed to learn a deep feature representation in which distribution shifts are reduced.

2.2. Convolutional layers and pooling layers

One-dimensional (1D) CNN has high efficiency with characteristics like parameter sharing and sparse connecting. In this study, 1D CNN is utilized to obtain abstract nonlinear representations of the input data and encode local spatial information by convolutional filters. As shown in Fig. 1 1D CNN consists of convolutional layers and max-pooling layers. According to the filter length, the convolutional layer extracts local sequences from input data. Then filter kernels convolve with local subsequences and generate convolutional feature maps with an activation function. Each filter applies a similar transformation to each subsequence. The convolution operation can be described as:

$$v_j^{L+1} = \varphi((b_j^{L+1} + \sum_i K_{ij}^{L+1} * v_i^L)) \quad (1)$$

where v_i^L represents the output of i^{th} channel in the layer L , φ represents the activation function, adopted Rectified Linear Unit (ReLU) function, b_j^{L+1} and K_{ij}^{L+1} respectively represent the corresponding bias vector and weight, $*$ denotes the convolution operator, and L represents the number of kernels.

Then, the max-pooling layer performs the downsampling step of the output convolutional representation by taking the maximum value over the window defined by pool size. Besides, the global average pooling (GAP) technique is popular to vectorize input tensors by outputting the expectation of every feature map [14], which interprets feature maps as categories confidence maps [29]. Therefore, GAP squeezes the spatial dimension where all regions contribute to the output. It is also famous for no parameter to optimize, which avoids overfitting and improves the robustness of spatial translations. This paper mainly utilizes GAP as a struc-

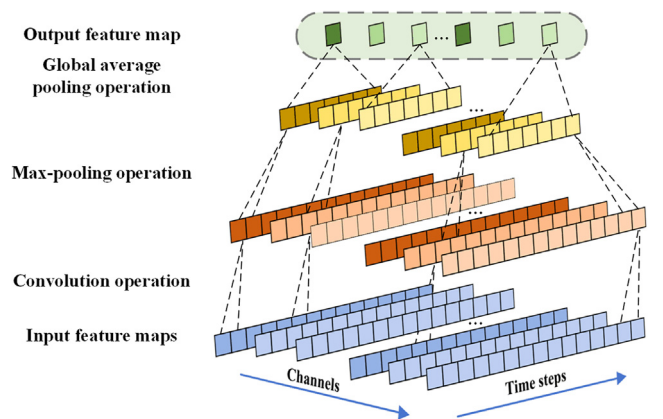


Fig. 1. The illustration of convolution and pooling operation.

tural regularizer that relates feature maps to confidence maps of channels while reducing the dimension.

2.3. Hilbert space embedding

For a random variable X with domain D and corresponding instantiation x , Hilbert space embedding represents its distribution $P(X)$ by an element in a reproducing kernel Hilbert space $\mathcal{H}$. A reproducing kernel Hilbert space $\mathcal{H}$ on D with a kernel k is a Hilbert space of function $f : D \rightarrow \mathbb{R}$. Its inner product $\langle \cdot \rangle_H$ owns the reproducing properties:

$$\langle f(\cdot), k(x, \cdot) \rangle_H = f(x) \quad (2)$$

$$\langle k(x, \cdot), k(x', \cdot) \rangle_H = k(x, x') \quad (3)$$

where a function f can be evaluated at $x \in X$ as an inner product.

Alternatively, $k(x, \cdot)$ can be regarded as an infinite-dimensional feature map $z(x)$:

$$\langle z(x), z(x') \rangle_H = k(x, x') \quad (4)$$

Given $\mathbb{E}_X[k(X, X)] < \infty$, the probability distribution $P(X)$ is mapped to the expected feature map, i.e., endowed to an element in $\mathcal{H}$ by kernel embedding with k [30,31]:

$$\mu_X(P) \triangleq \mathbb{E}_X[z(X)] = \int_D z(x) dP(x). \quad (5)$$

3. The proposed method

This section presents a novel joint adaptive transfer learning network based on multi-layer feature fusion to strengthen the learning capability with a small amount of labeled data, as shown in Fig. 2. First, the MJDCNN is established based on the multilinear

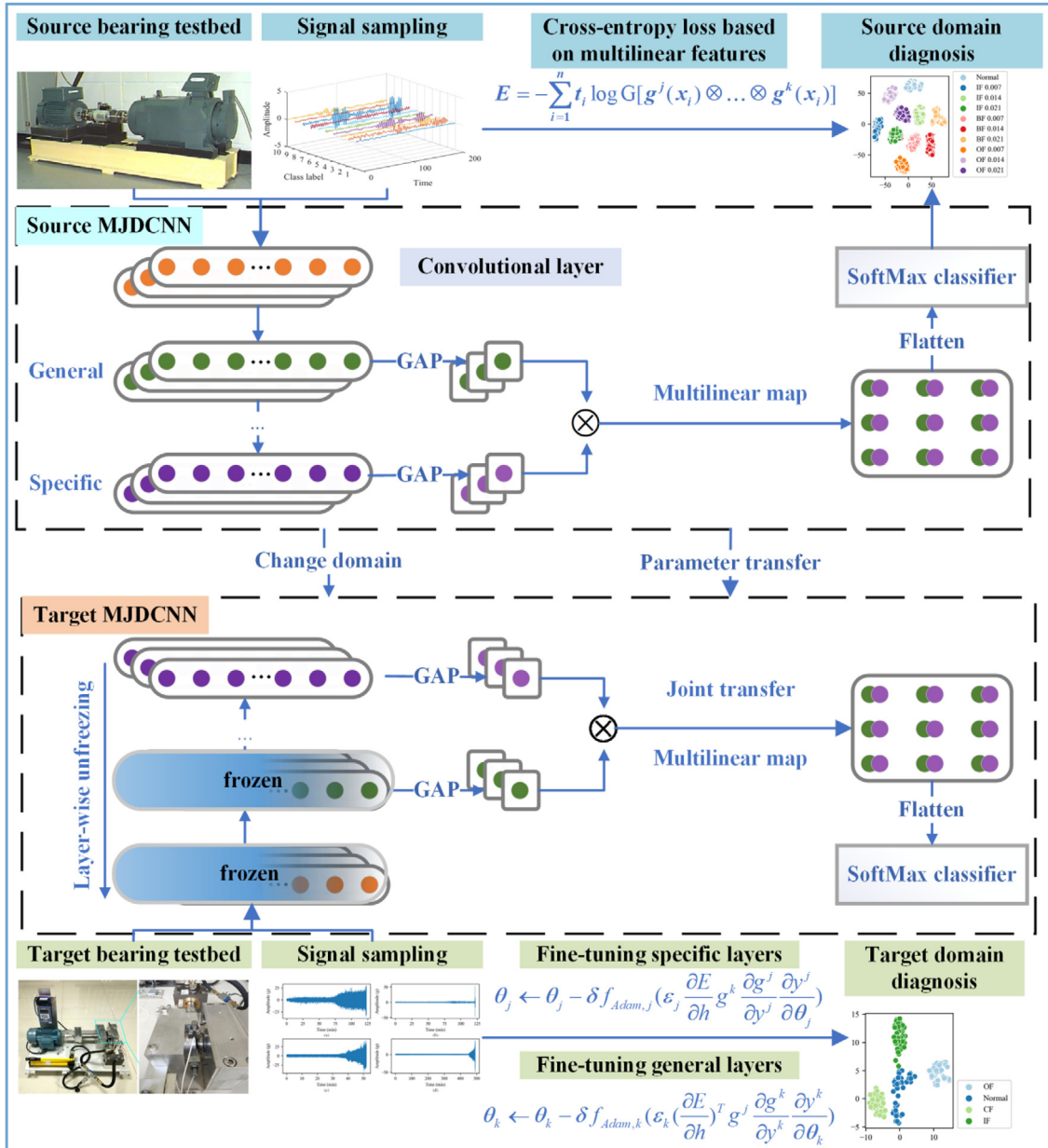


Fig. 2. The proposed MJDCNN under the JATL framework.

map. Next, the JATL framework is proposed to exploit cross-domain-invariant information conveyed in the bottom-layer features to assist cross-domain adaptation. Ultimately, the overall procedure and contributions of the proposed framework are summarized.

3.1. The architecture of MJDCNN

It is noted that response neurons at different depths are sensitive to varying abstract information [32]. At the same time, it is proved that the asymmetric network structure and feature fusion between layers facilitate solving vanishing gradient and representation bottleneck [29]. Motivated by the above considerations, we proposed MJDCNN to achieve unique feature fusion based on a multilinear map.

MJDCNN consists of a set of feature extractor layers $F^i_{(i=1,2,\dots)}$, global average pooling layers $GAP^i_{(i=1,2,\dots)}$, a mapping layer M, and an output layer G. F^i consists of a hierarchy of convolutional and pooling layers, which is defined as

$$F^i: x \rightarrow y^i \in \mathbb{R}^{w^i \times c^i} \quad (6)$$

where $x \in \mathbb{R}^{W \times C}$ represents samples of the input signal with C length and W signal channels. Through F^i , the input signal is transformed into convolutional feature maps y^i with c^i length and w^i channels. The output of GAP^i is the expectation of feature maps on channels, which is defined as

$$GAP^i: y^i \rightarrow g^i \in \mathbb{R}^{w^i \times 1} \quad (7)$$

where g^i is obtained with kernel k^i in reproducing kernel Hilbert space $\mathcal{H}^i$ for y^i .

MJDCNN processes the signal with an arbitrary length in a forward-propagation step with feature extractor layers and GAP layers. The mapping layer M embeds the joint distribution $C_h(P)$ of multiple variables $g^1, \dots, g^k$ into tensor product feature space $\mathcal{H}^1 \otimes \dots \otimes \mathcal{H}^k$ using a multilinear map [30]. It is noted that the empirical estimate of $C_h(P)$ by finite samples can converge to its population distribution as well [33]. The embedding of $C_h(P)$ into infinite-dimensional feature space can completely preserve statistical features, which can be formulated as the outer product of multiple random vectors:

$$M: g^1 \otimes \dots \otimes g^k \rightarrow h \quad (8)$$

where if all the variables except g^k keep constant, then h is a linear function of g^k .

After reshaping the multilinear feature map to a 1D vector, a signal descriptor is obtained. It is used for the output layer G which utilizes the SoftMax function to return the probability that the output belongs to each health state. And the cross-entropy loss of input samples $x = \{x_i\}_{i=1}^n$ and labels $t = \{t_i\}_{i=1}^n$ is applied for evaluating errors between predictive output and true labels as

$$E = - \sum_{i=1}^n t_i \log G[g^1(x_i) \otimes \dots \otimes g^k(x_i)] \quad (9)$$

where $G[g^1(x_i) \otimes \dots \otimes g^k(x_i)]$ represents the i^{th} multilinear output. Using backpropagating errors, the weights of model parameters will be updated continuously, and the mapping relationship between joint embeddings and health states is established.

The input feature maps of a mapping layer must have a compatible dimension. The proposed approach provides a flexible and effective feature fusion. As the feature maps are vectorized into an expected feature map by the GAP layer, a compatible dimension

for the multilinear map is always available even without the same cardinalities. Accordingly, this design helps MJDCNN become more practical than traditional multi-scale fusion. Moreover, as shown in Fig. 2, kernel embeddings provide a nonparametric approach to manipulate multi-layer feature fusions and their corresponding distributions without losing any statistical features [34]. Based on it, MJDCNN can capture the complex structures behind feature distribution effectively [33,35]. What's more, a dropout layer is used as a regularization technique with a rate of 0.5. The Early Stopping is set to terminate the training process when the observed validation loss is no longer decreasing within 60 consecutive epochs. Before the training process, 20% of training data are randomly split into validation data.

3.2. Joint adaptive transfer learning framework

MJDCNN is effective for extracting discriminative features and realizing multi-layer feature fusion. However, the collected data are usually insufficient to train reliable cross-domain diagnosis models. Consequently, an approach capable of reliably transferring knowledge and reducing cross-domain distribution discrepancy is urgently needed. To ensure general information can prospectively assist the domain adaption without being distorted by high-level specific layers and capture the cross-domain distribution shift [33], the joint distribution of the feature maps with different transferability is utilized. Based on it, a two-stage JATL framework is devised, as shown in Fig. 2. First, an MJDCNN is constructed and trained on the source domain for feature learning in the pre-training stage. From a high-level layer j and a low-level layer k with an input x , the convolutional feature maps $y^j = F^j(x)$ and $y^k = F^k(x)$ are obtained, respectively. Then, with $g^j = GAP^j(y^j)$ of size $w_j \times 1$ and $g^k = GAP^k(y^k)$ of size $w_k \times 1$, the multilinear features $h(g^j, g^k) = g^j (g^k)^T$ of size $w_j \times w_k$ are obtained. Since the architecture of MJDCNN is a directed acyclic graph, the parameters θ in this model are optimized by back-propagating in an end-to-end way with Eq. (9):

$$\hat{\theta} = \arg \min_{\theta} E(x, t) \quad (10)$$

where $\hat{\theta}$ denotes the optimal value of θ .

Specifically, let $\partial E / \partial h$ be the gradients of Eq. (9), and based on the chain rule we have:

$$\frac{\partial E}{\partial g^j} = \left(\frac{\partial E}{\partial h} \right) g^k, \quad (11)$$

$$\frac{\partial E}{\partial g^k} = \left(\frac{\partial E}{\partial h} \right)^T g^j. \quad (12)$$

Then, in the transfer learning stage, a reliable transfer process is achieved with second-order feature descriptors $h(g^j, g^k)$. The network is tuned to optimize Eq. using Adam with mini-batches. According to layer-wise quantitative researches, g^k is more likely to be cross-domain-invariant which is instructional to the transfer process for higher layers, while g^j are prone to be not safely transferable [27]. The layers are layer-wise unfrozen and fine-tuned in the JATL process with a small amount of annotated data. The bottom layers will be slightly fine-tuned after the higher layers are tuned to convergence. When F^k is frozen, only layers with parameters θ_j after the layer k are updated with Eq. (11) as

$$\begin{aligned} \theta_j &\leftarrow \theta_j - \delta f_{Adam,j}(\epsilon_j \frac{\partial E}{\partial \theta_j}) = \theta_j - \delta f_{Adam,j}(\epsilon_j \frac{\partial E}{\partial g^j} \frac{\partial g^j}{\partial \theta_j}) \\ &= \theta_j - \delta f_{Adam,j}(\epsilon_j \frac{\partial E}{\partial h} g^k \frac{\partial g^j}{\partial \theta_j}) \end{aligned} \quad (13)$$

where δ represents the learning rate. $f_{Adam,j}$ represents the computation function in the Adam optimizer for θ_j . ε_j represents the penalty coefficient.

Moreover, when F^k is unfrozen, layers with parameters θ_k before the layer k are updated with Eq. (12):

$$\begin{aligned}\theta_k &\leftarrow \theta_k - \delta f_{Adam,k}(\varepsilon_k \frac{\partial E}{\partial \theta_k}) = \theta_k - \delta f_{Adam,k}(\varepsilon_k (\frac{\partial E}{\partial g^k} \frac{\partial g^k}{\partial y^k} \frac{\partial y^k}{\partial \theta_k})) \\ &= \theta_k - \delta f_{Adam,k}(\varepsilon_k (\frac{\partial E}{\partial h} \frac{\partial h}{\partial g^k} \frac{\partial g^k}{\partial y^k} \frac{\partial y^k}{\partial \theta_k}))\end{aligned}\quad (14)$$

where $f_{Adam,k}$ represents the computation function in the Adam optimizer for θ_k . ε_k represents the penalty coefficient.

According to Eq. (13) and (14), θ_j and θ_k use using features from the other layer for upgrading, respectively. It illustrates how the JATL framework implements mature cross-domain joint adaptation. During the backpropagation, the multilinear map decoupling g^j and g^k enables the transfer of gradients directly into the bottom layer without nonlinear transformation, ensuring a stable and easier training process. Unfrozen layers will be fine-tuned together until the validation loss reaches the Early Stopping requirements or the defaulted maximum iterations. Finally, the transferred MJDCNN is evaluated against the fault diagnosis task on the target domain. What's more, similar to the quadratic kernel expansion, multilinear features $h(g^j, g^k)$ allow g^j and g^k to be conditioned on each other when their interactions are considered [34]. Motivated by it, firstly, the low-level F^k is frozen without tuning and conditioned on when adaptively tuning the high-level F^j . And the mean map $\mathbb{E}_h[g_j \otimes g_k]$ can be rewritten in the way of conditional embedding [36]:

$$\mathbb{E}_h[g_j \otimes g_k] = \mathbb{E}_{y^k}[\mathbb{E}_{y^j|y^k}[g^j|y^k] \otimes g^k]. \quad (15)$$

This mean map $\mathbb{E}_h[g_j \otimes g_k]$ shows that a carry track connecting layers with different transferability is effectively created by keeping y^k as reliable prior information. Consequently, employing a fused representation of features with various transferability, the proposed JATL framework makes the utmost of learned domain-invariant information to achieve better transferability and cross-domain adaptation.

3.3. General procedure of the proposed transfer learning framework

This paper presents a novel joint adaptive transfer learning framework for the cross-domain diagnosis of rotating machines. The general procedure can be divided into five steps:

Step 1: Sufficient labeled data on the source domain and limited labeled data in the target domain are collected by sensors installed on running machines.

Step 2: Both source and target data are normalized to be within [0,1]. Then, the preprocessed data are divided into the training and testing datasets. The testing dataset is just used to evaluate the performance of the model without joining the training process.

Step 3: MJDCNN is constructed and trained on the source dataset in the pre-training stage while its parameter weights are stored.

Step 4: In the transfer learning stage, this pre-trained model is used to initialize the model on the new task and is fine-tuned using the JATL framework with a small number of labeled target domain samples. If the new task has the same label space as the source task, the MJDCNN is fully frozen, then layer-wise unfrozen and fine-tuned. Otherwise, the output layer will be modified to adapt to the new task with a different label space. Furthermore, the layer below the low-level one used in the multilinear map is frozen in MJDCNN and then is layer-wise unfrozen and fine-tuned.

Step 5: The testing accuracy of our proposed framework is evaluated on the testing dataset on the target domain.

The contributions of this study can be concluded as:

- 1) *A novel feature fusion makes better use of hierarchical information.* MJDCNN embeds the multi-layer joint distribution into the reproducing kernel Hilbert space to flexibly fuse multi-layer features based on the multilinear map and the designed network architecture. By combining lossless early statistics information and higher abstract features, MJDCNN can significantly improve feature representation capability in a nonparametric approach.
- 2) *An excellent-performance joint adaptive transfer learning framework.* The joint adaptive transfer learning framework both considers the domain adaptation needing for the entire model and the transferable ability at different layers. Upon obtaining general features from the bottom layers, the transfer process of the higher layer is instructed accordingly. This novel transfer learning model copes with cross-domain fault diagnosis tasks lacking sufficient training data. The comparative experiments have proved the superiority of devised transfer learning framework over traditional models.
- 3) *An end-to-end intelligent fault diagnosis approach with high prediction accuracy.* Based on stacked 1D CNNs, this approach has achieved high diagnosis performance and generalization processing raw 1D raw vibration data directly. Without high dependence on handcrafted features, applying this IFD model to practical engineering scenarios is promising.

4. Experimental study

Three experiments are conducted on two benchmark datasets to evaluate the proposed MJDCNN and JATL framework. Details of these experiments, including training and testing procedures, benchmark models, and evaluation criteria, are provided. All experiments have been conducted on a computer with RYZEN_7_3700X CPU, 16 GB RAM, and a NVIDIA RTX 2070 GPU. The level of difficulty for these experimental scenarios is increased progressively. Experiment I evaluates the effectiveness of the proposed MJDCNN on a mixture condition-based dataset. Experiment II verifies the diagnostic knowledge transfer capability of the proposed JATL framework under varied working conditions on the same machine. Experiment III is a more challenging situation, namely cross-machine diagnosis, involving complicated factors, such as diverse machine specifications [6].

4.1. Case 1: Experiments on CWRU dataset

4.1.1. Dataset description

In Case 1, raw experimental bearing data are provided by the Case Western Reserve University (CWRU) Bearing Data Center, and the testing platform and apparatus are shown in Fig. 3 [37]. Under four kinds of working conditions (motor loads of 0 to 3 HP), vibration data of motor bearings are documented. There are three fault types: inner raceway fault (IF), ball element fault (BF), and outer raceway fault (OF). And on each kind of fault location, there are three fault levels according to the crack diameter of 0.007, 0.014, and 0.021 in.. Thus, health states include one normal state and nine fault states. The data are collected using accelerometers located at the drive end of the motor housing, and the sampling frequency is 12 kHz.

4.1.2. Experiment setup

In designed experiments, each sample contains 400 data points. Each health status contains 300 samples under each working condition, and a total of 3000 samples are collected for each condition. As shown in Table 1, the bearing dataset is divided into subsets on the source and target domains. Subsets $S_0 - S_3$ under working con-

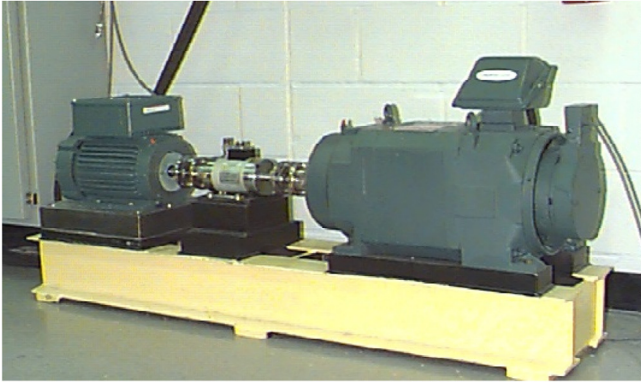


Fig. 3. The test platform for CWRU bearing dataset.

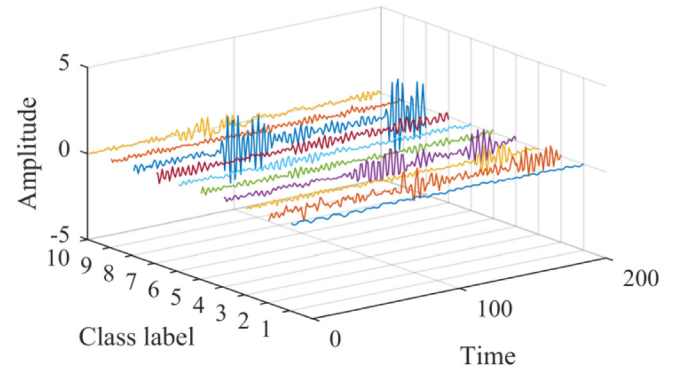


Fig. 4. Segments of vibration signals from working condition (1) in the CWRU dataset.

ditions 0–3 HP are taken as being drawn from the source domain. Each health state contains 200 samples as training data and 100 samples as testing data in $S_0 - S_3$. As a contrast, subsets $T_0 - T_3$ are taken as being drawn from the target domain. They contain 40 samples from each fault state as training data and the left 260 samples as testing data. Some segments of vibration signals of different health states under working condition (1) are shown in Fig. 4. The x-axis represents time, the y-axis represents the ten kinds of health states, while the vertical axis represents the amplitude of vibration signals.

With the above samples, we explore the performances of the proposed approach from two levels. Firstly, to evaluate the performance of the proposed MJDCNN, Experiment I is conducted on a mixture of subsets $S_0 - S_3$, which contains 8000 training samples and 4000 testing samples where this mixture of data from four kinds of working conditions is widely adopted to validate proposed new models [22,38,39]. Then, to evaluate the superiority of the proposed JATL framework on cross-condition tasks, Experiment II is conducted by comparing it with other advanced models and strategies. The model is pre-trained on training samples of $S_0 - S_3$ and then transferred to the fault diagnosis task $T_0 - T_3$ under other working conditions.

4.2. Case 2: Experiments on XJTU-SY dataset

4.2.1. Dataset description

In Case 2, the effectiveness of the proposed approach is studied in a cross-machine fault diagnosis scenario where target data subsets are constructed from XJTU-SY bearing dataset [40]. As shown in Fig. 5, accelerated degradation signals of LDK UER204 bearings are collected under different operating conditions. Two PCB 352C33 accelerometers are mounted to collect vibration signals

on the housing of test bearings. Sampling frequency and period equal to 25.6 kHz and 1 min, respectively. It has been illustrated that vibration signals in the horizontal direction have more abundant degradation information [40]. Therefore, the horizontal vibration signals are chosen for cross-machine diagnosis.

Fig. 6 illustrates that as the run-to-failure process progresses, the vibration signal continues to grow in amplitude. The point when tested bearings begin to deteriorate is detected based on the adaptive degradation detection method using the RMS- 3σ criterion [41]. There are five kinds of health conditions in this dataset: normal state, outer raceway fault (OF), inner raceway fault (IF), composite fault of OF and IF (OI), and cage fault (CF). Compared to Table 1, this dataset contains new fault types such as composite fault (OI) and cage fault (CF). On top of that, unlike the CWRU dataset, which introduces simple single-point faults with specific diameters by electro-discharge machining, degradation signals throughout the whole operating life of the XJTU-SY dataset are more complicated. Even with the same fault location as OF and IF, the fault diameters are different.

4.2.2. Experiment setup

As shown in Table 2, to further evaluate the transferability of the proposed approach on varying label spaces and unknown label information, Experiment III is conducted to transfer pre-trained MJDCNNs on $S_0 - S_3$ to cross-machine new target subsets $T_4 - T_5$.

4.3. MJDCNN architecture and diagnosis performance metrics

The parameter setting and architecture of MJDCNN are shown in Table 3. Con represents the standard convolutional layer, and MP stands for the max-pooling layer. The multilinear map of outputs from the second and the last convolutional layer is chosen

Table 1
Details of the experimental dataset based on CWRU bearing dataset.

Working Condition	Fault Types	Fault Diameters	Number of Samples (training/testing)		Category Labels
			$S_0 - S_3$	$T_0 - T_3$	
0–3	Normal	0	200/100	40/260	1
0–3	IF	0.007	200/100	40/260	2
0–3	IF	0.014	200/100	40/260	3
0–3	IF	0.021	200/100	40/260	4
0–3	BF	0.007	200/100	40/260	5
0–3	BF	0.014	200/100	40/260	6
0–3	BF	0.021	200/100	40/260	7
0–3	OF	0.007	200/100	40/260	8
0–3	OF	0.014	200/100	40/260	9
0–3	OF	0.021	200/100	40/260	10

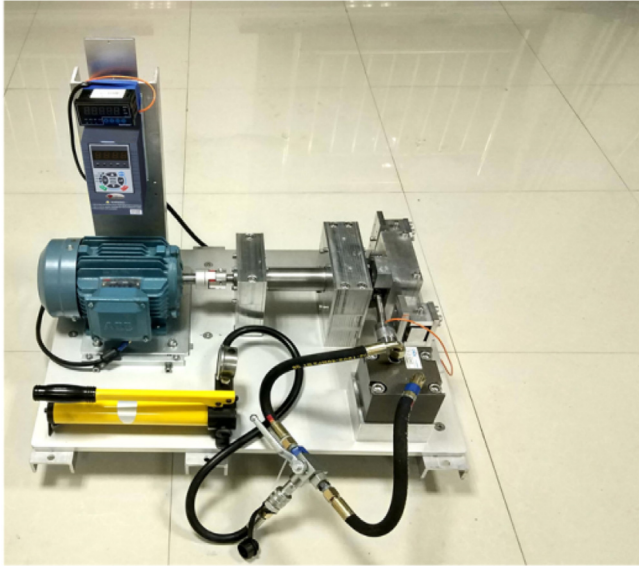


Fig. 5. The testbed of the XJTU-SY bearing dataset.

to get fused features by the mapping layer. The features at the second layer have less noise and are relatively general, while at the final layer are specific to the ultimate diagnosis task. And the fused features are vectorized for fault pattern recognition directly without being destroyed by intermedia layers. The grid search algorithm is utilized to determine the optimal hyper-parameters,

Table 3
Parameter setting of MJDCNN.

Layer Name	Kernel Size/Number	Stride	Connected to
Con ¹	25/8	1	Input
MP ¹	8/8	8	Con ¹
Con ²	3/16	1	MP ¹
Con ³	8/18	1	Con ²
MP ²	2/18	2	Con ³
Con ⁴	5/32	1	MP ²
MP ³	2/64	2	Con ⁴
GAP ¹	/	/	Con ²
GAP ²	/	/	Con ⁴
Mapping	/	/	GAP ¹ GAP ²
Flatten	/	/	Mapping
Dropout	/	/	Flatten
SoftMax	10/	/	Dropout

including learning rate, etc. The proposed MJDCNN architecture comprises four convolutional and three max-pooling layers, two GAP layers, and finally, an output layer. The convolutional kernels are selected relatively small to prevent losing local features, except those in the first layer, which are somewhat wide for suppressing high-frequency noise [16]. The number of convolutional filters increases as the convolutional layers get deeper. The batch size is 128 and 16 for pre-training and fine-tuning, respectively. The max training epoch is determined as 500, and the initial learning rate is set as 0.001. The exponential decay rate of the first moment of Adam is set as 0.9, and the exponential decay rate of the second moment is set as 0.999.

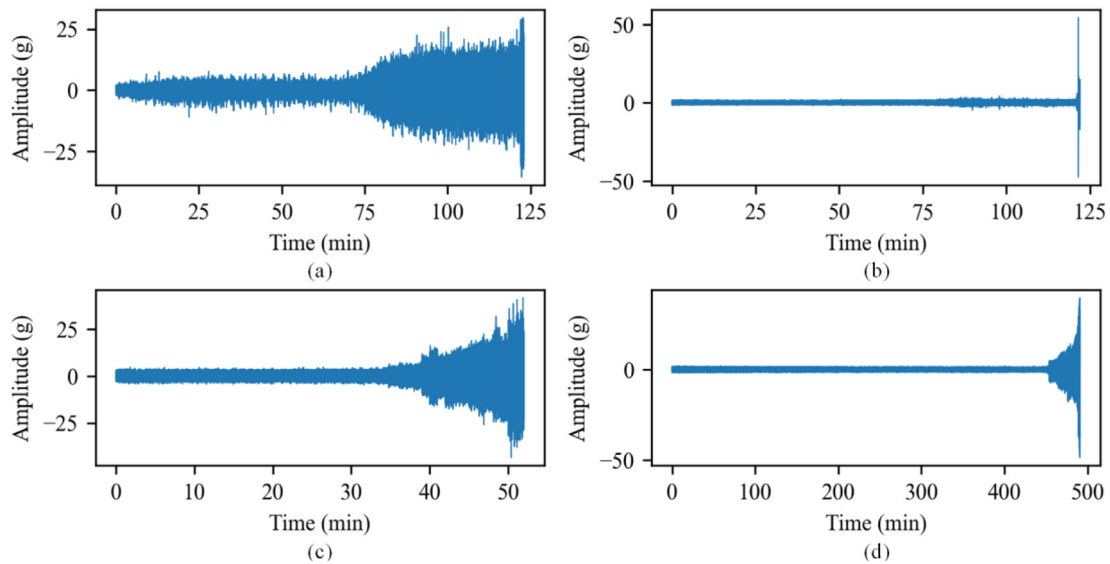


Fig. 6. Vibration signal of a bearing with (a) OF; (b) IF; (c) OI and (d) CF.

Table 2
Details of the experimental dataset based on the XJTU-SY bearing dataset.

Working Condition	Target Dataset	Fault Types	Bearings	Number of Samples (training/testing)	Category Labels
1	T_4	Normal	1_1, 1_2, 1_3, 1_4, 1_5	40/260	1
		OF	1_1, 1_2, 1_3	40/260	2
		CF	1_4	40/260	3
		OI	1_5	40/260	4
		IF	1_1	40/260	5
2	T_5	Normal	2_1, 2_2, 2_3, 2_4, 2_5	40/260	1
		OF	2_2, 2_4, 2_5	40/260	2
		CF	2_3	40/260	3
		IF	2_1	40/260	5
		OI	2_5	40/260	4

After shuffling samples randomly, all models are trained repeatedly ten times to avoid the influence of random fluctuation. The average classification accuracy evaluates performances in all experiments. What's more, the precision ratio and recall ratio are adopted to quantitatively assess the transfer efficiency in the way of the P-R curve, which are formulated as follows:

$$\text{Precision ratio} = \frac{TP}{TP + FP} \quad (16)$$

$$\text{Recall ratio} = \frac{TP}{TP + FN} \quad (17)$$

where TP , FP and FN represent the number of true positive instances, false-positive instances, and false-negative instances, respectively.

5. Results and discussions

5.1. Experiment I: Mixture condition-based effectiveness evaluation

In Experiment I, the proposed MJDCNN model is evaluated on mixed samples from four kinds of working conditions. As a result, the average testing accuracy is 99.95% (3998/4000) which is exceptionally high. The feature fusion makes full use of the advantages of different layers and breaks the representation bottleneck of traditional models.

To further explore the internal transformation and learned representation between the continuous layers in the MJDCNN, we visualize its intermediate activations using t-SNE technology, as shown in Fig. 7. The degree of decentralized distribution phenomenon between different categories is increasing as the layer goes deeper. And surprisingly, as shown from Fig. 7(e) to (f), although no more dense connection layer or convolutional layer is added, more discriminative features are obtained effectively by the multilinear map of outputs from different layers. By introducing earlier statistical information into the final diagnosis, the clus-

Table 4

Classification accuracy of varying approaches on the diagnosis tasks under 0 HP.

Approach	$S_1 \rightarrow T_0$	$S_2 \rightarrow T_0$	$S_3 \rightarrow T_0$	Average
Proposed JATL	97.54%	97.58%	95.96%	97.03%
N1	94.50%	96.90%	85.80%	92.40%
N2	95.88%	95.54%	94.92%	95.45%
MSCNN	95.50%	95.42%	93.31%	94.74%
TCNN	88.92%	94.27%	88.38%	90.52%
CNN_Li	87.19%	89.58%	88.04%	88.27%
VGG-16	91.23%	91.23%	91.23%	91.23%

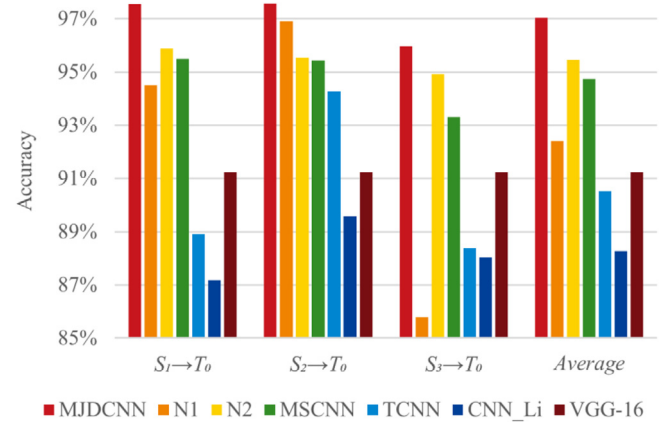


Fig. 8. Diagnosis accuracy comparison results on target domain 0 HP.

ter becomes more concentrated, better distinguished, and most categories are separable with little overlap in the mapping layer. It indicates that this multilinear fusion is tailored to learn discriminative features and enable a simpler way for fault pattern recognition. Based on Experiment I, the effectiveness of the proposed

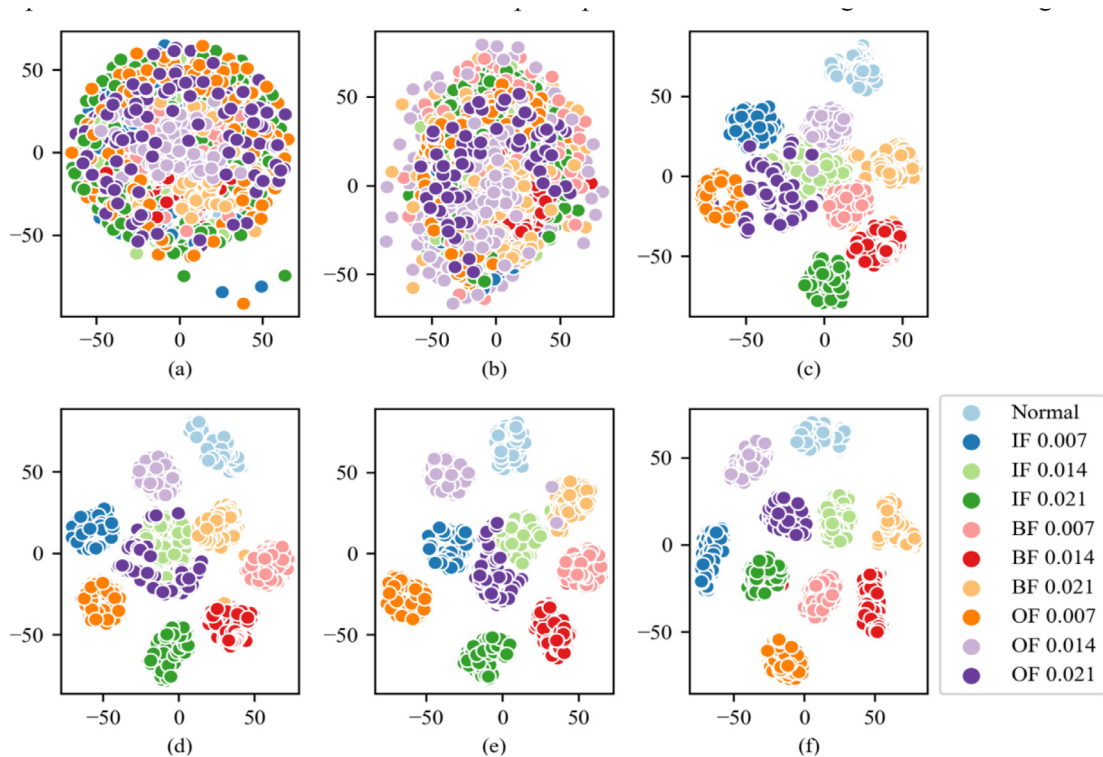
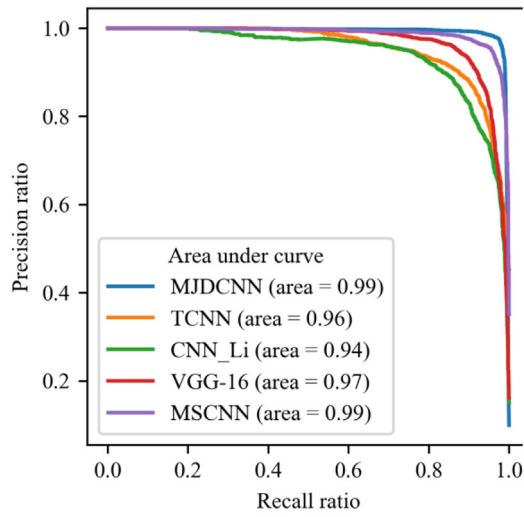
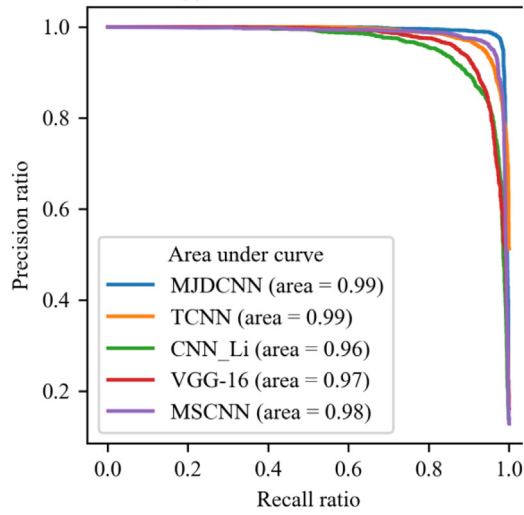


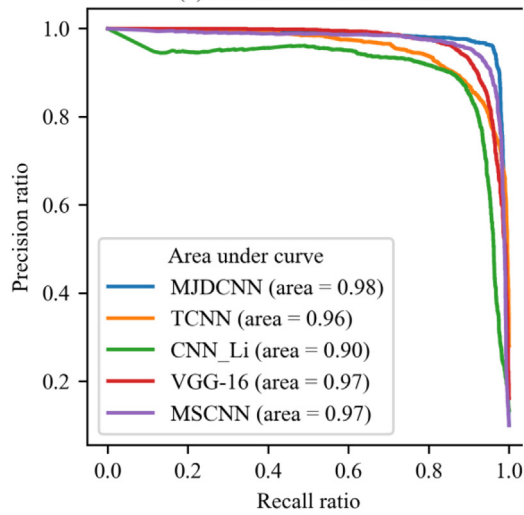
Fig. 7. Feature visualization for activations in MJDCNN from different layers via t-SNE: (a) Input; (b) Con¹; (c) Con²; (d) Con³; (e) Con⁴; (f) Mapping.



(a) P-R curve based on 1 HP.



(b) P-R curve based on 2 HP.



(c) P-R curve based on 3 HP.

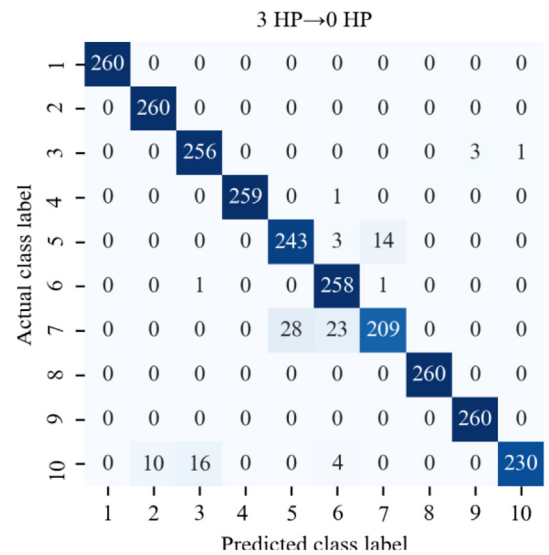
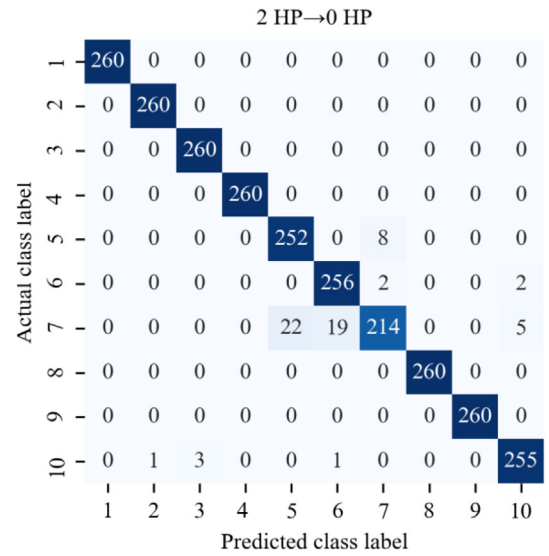
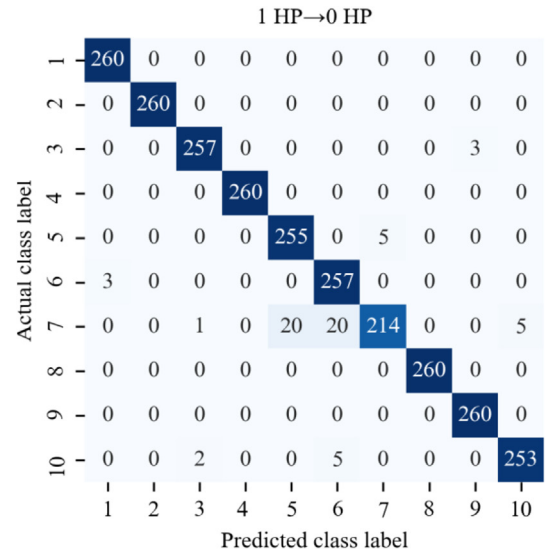


Fig. 9. P-R curve comparison results on target domain 0 HP.

Fig. 10. The confusion matrixes of different transfer processes.

MJDCNN is verified and ensures a prerequisite for the following transfer learning.

5.2. Experiment II: Effect of the proposed approach based on variable working conditions

Experiment II is conducted to demonstrate the feasibility and necessity of applying the proposed JATL framework in small sample scenarios. Previous studies proved that the diagnosis tasks under 0 HP with limited samples are relatively complicated [16]. Thus, as shown in Table 4, we design three transfer processes to explore the feasibility of our approach. Notably, in our scenario, the amount of training data is much less than that in previous studies. $S_i \rightarrow T_j$ is defined as pre-training a diagnosis model with the source subset S_i and fine-tuning it with the target subset T_j . The results are firstly compared with two baseline models, and their transfer learning strategies are described as follows to investigate the superiority of our proposed approach:

Table 5
Diagnosis performance of the proposed approach on varying transfer paths.

Source subset	Target subset			
	T_0	T_1	T_2	T_3
S_0	/	98.38%	99.04%	99.15%
S_1	97.54%	/	99.81%	99.81%
S_2	97.58%	99.23%	/	99.92%
S_3	95.96%	98.58%	99.88%	/

Table 6
Diagnosis accuracy of different approaches on the XJTU-SY dataset.

Target subset	Approach	Cross-machine source subset (CWRU)					Subset with sufficient samples (XJTU-SY)	
		S_0	S_1	S_2	S_3	Average	S_4	S_5
T_4	MJDCNN	93.94%	94.71%	94.23%	92.50%	93.85%	96.73%	/
	MSCNN	88.65%	85.38%	88.65%	86.83%	87.38%	94.81%	/
	TCNN	87.31%	93.37%	93.56%	92.40%	91.66%	94.71%	/
	CNN_Li	72.79%	82.50%	65.96%	74.81%	74.02%	90.29%	/
T_5	MJDCNN	91.73%	88.56%	87.50%	91.25%	89.76%	/	96.06%
	MSCNN	77.88%	77.79%	80.67%	77.31%	78.41%	/	95.48%
	TCNN	78.56%	86.15%	86.73%	88.46%	84.98%	/	87.12%
	CNN_Li	65.48%	62.12%	62.31%	60.10%	62.50%	/	70.10%

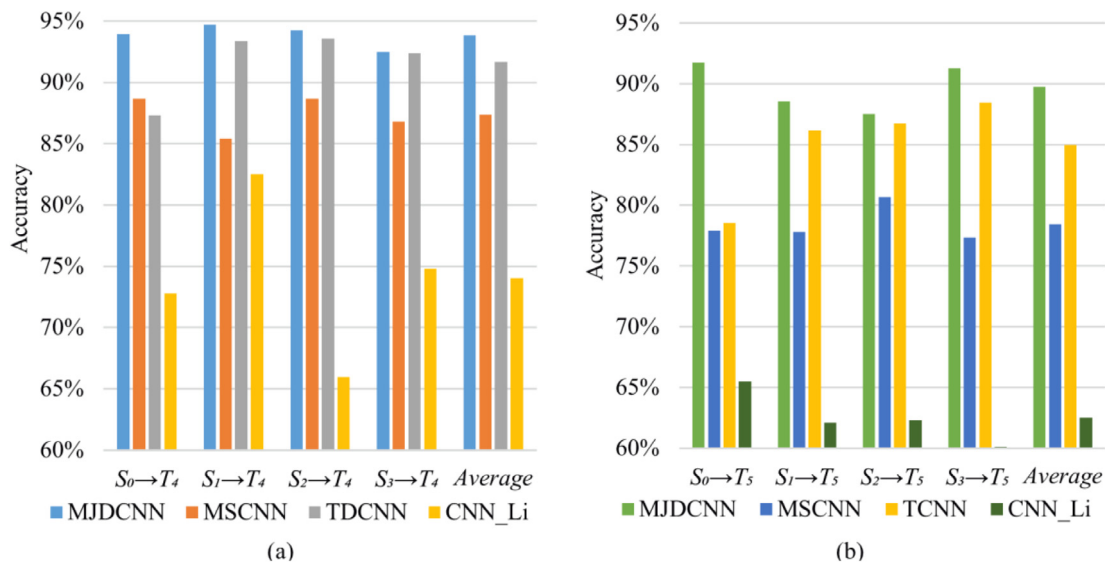


Fig. 11. Diagnosis accuracy comparison results on (a) T_4 and (b) T_5 .

N1: MJDCNN model without JATL process.

N2: A sequential CNN model whose backbone structure is the same as MJDCNN but without mapping and GAP layers. In transferring part, the layer-wise approach is also used for domain adaption of N2.

N1 is used to illustrate the effectiveness of transfer learning. And the comparison with N2 is to show the influence of the JATL framework. The proposed approach is also compared with state-of-the-art transfer learning approaches, i.e. TCNN, MSCNN, CNN_Li, VGG-16 [16,21–23]. The first three approaches employ the layer-wise fine-tuning strategy. And for the VGG-16, the raw vibration data are first transformed into frequency-domain images using Fast Fourier transformation [42]. And the VGG-16 is pre-trained on ImageNet, and then all layers are fine-tuned except the bottom three blocks to adapt target tasks [22].

As shown in Fig. 8, with the increase of domain shift, the performance of N1 drops more seriously and gets the worst effects in the process $S_3 \rightarrow T_0$. Correspondingly, the diagnosis accuracy of MJDCNN under the proposed JATL is 3.22%, 0.7%, 12.57% higher than that of N1 under three transfer processes, respectively. What's more, although N2, the sequential model, maintains a certain accuracy, the lack of sufficient cross-domain-invariant knowledge to assist higher-level adaption leads to less effective results. On the three processes, the accuracy of the proposed approach achieved is 1.73%, 2.14%, 1.10% higher than that of N2, respectively. Fusing more general features directly, the results prove that the transfer process of JATL is more reliable.

Besides, comparing the results of other advanced transfer learning methods shows the superiority of the proposed framework. As shown in Fig. 9, during these transfer processes, the P-R curves of the other four models are entirely covered by that of MJDCNN using the JATL framework. And for the area under curve (AUC), the proposed approach is also more prominent than others. Furthermore, compared with MSCNN that increases feature dimensions but retains the information under each channel, the amount of information in each dimension increases using the multilinear map. The proposed JATL framework brings in interactive knowledge, which facilitates improving the diagnosis performance.

Moreover, the multi-classification confusion matrixes of the proposed JATL on different transfer processes to target domain 0 HP are presented in Fig. 10. The elements on the main diagonal of the matrix represent the properly diagnosed number of each fault state. It can be observed that even with a minority amount of labeled new data, misleading diagnosis results are relatively low. What's more, the diagnosis performance of the proposed

approach on other transfer paths under varying working conditions is presented in Table 5. It is a common scenario in real engineering practice that a change in working conditions is often inevitable. This phenomenon demands a robust transfer learning approach. As shown in Table 5, the transfer aiming at other target domains shows more stable and better diagnosis performance, which illustrates that diagnosis under 0 HP is indeed relatively challenging. Furthermore, with an accuracy of over 98% for most diagnosis tasks, achieving high-quality IFD in real practice is promising even with a small amount of labeled data.

5.3. Experiment III: Effect of the proposed approach based on another machine

In Experiment III, MJDCNN pre-trained under $S_0 - S_3$ is transferred to $T_4 - T_5$ respectively using the JATL framework. To illustrate the influence of small samples and the corresponding processing capacity of the proposed approach, two training subsets $S_4 - S_5$ contained 200 training samples of each health state under these two working conditions are constructed. $S_4 \rightarrow T_4$ and $S_5 \rightarrow T_5$ are generally supervised deep learning applications with sufficient data and without distribution discrepancy. Moreover, other end-to-end approaches, MSCNN, TCNN, and CNN_Li, are also transferred to $T_4 - T_5$ for comparison. The comparison results are listed in Table 6 and Fig. 11.

For one thing, as these transfer learning diagnosis models are fine-tuned with small samples of a total of 160, they show relatively poorer performances compared to $S_4 \rightarrow T_4$ and $S_5 \rightarrow T_5$ which are trained with a total of 800 samples without distribution discrepancies. Fortunately, what stands out in Table 6 is the optimal accuracy of the MJDCNN fine-tuned with limited data. MJDCNN under the JATL framework still keeps a reliable level where $S_1 \rightarrow T_4$ is close to 95%. It indicates that the proposed approach has outstanding cross-machine transferability. Besides, the proposed framework can improve the diagnosis consistency with smaller performance variability across different transfer paths than the other networks. Notably, traditional transferred models display a more dramatic performance decline and greater variability on the XJTU-SY dataset. It indicates traditional models are sensitive to cross-machine discrepancies without making the utmost of cross-domain-invariant features.

What's more, it is noticeable that with numerous batch normalization layers for normalizing input samples, TCNN has achieved relatively better performance among these advanced models [21]. Nevertheless, due to a lack of adequate data, the performance

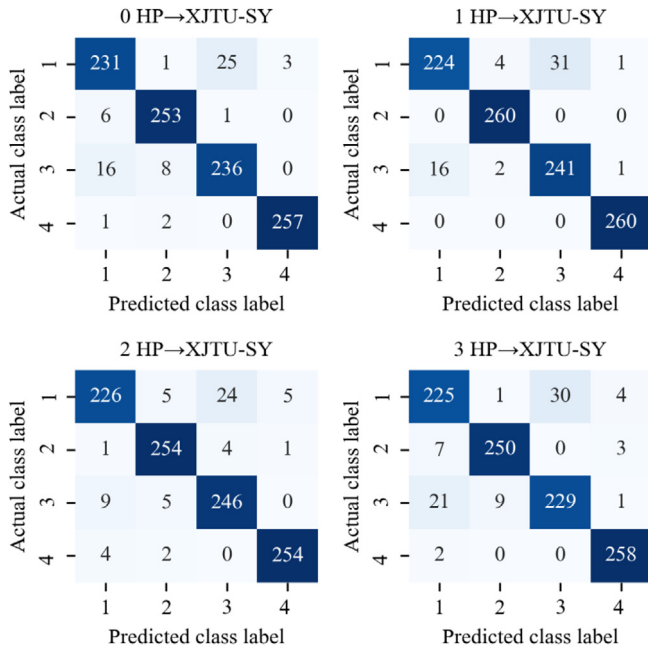


Fig. 12. The confusion matrixes of MJDCNN with different transfer paths to T_4 .

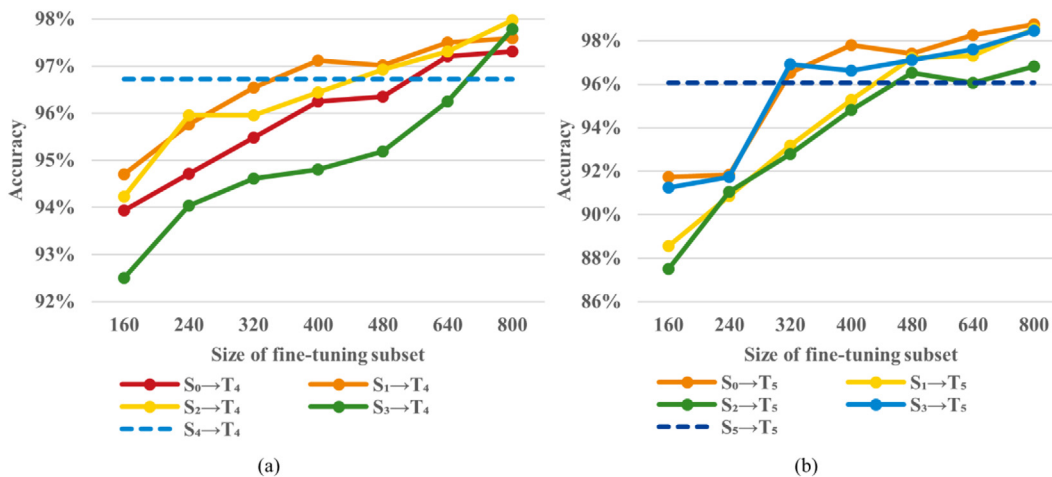


Fig. 13. Comparison results with different sizes of the fine-tuning subset on (a) T_4 , (b) T_5 .

of TCNN is limited. Overall, it can be inferred that even faced with the difficulty to train a reliable network for a small dataset, MJDCNN with the JATL framework is more general to a related cross-machine task. To further discuss the cross-machine transfer effect, four confusion matrixes on different transfer paths from different source subsets of CWRU to T_4 are shown in Fig. 12. Impressively, despite the array of new fault types, such as the cage fault (CF) and the composite fault (OI), the proposed transfer learning approach still provides a reliable diagnosis. It indicates that the proposed approach is more likely to develop a robust transfer learning network between different machines even with unknown label information.

For another thing, for reflecting the promotion effect for diagnosis, the performance of MJDCNN under the JATL framework with respect to different sample sizes for fine-tuning is investigated, as shown in Fig. 13. This figure exhibits that as the number of available training samples increases, the diagnosis accuracy of the proposed approach continuously improves. Besides, the proposed

approach breakthroughs the diagnosis performance of $S_4 \rightarrow T_4$ and $S_5 \rightarrow T_5$ even with a smaller amount of data than $S_4 - S_5$. The proposed JATL framework initializes MJDCNN with transferred knowledge from another machine producing a boost to generalization performance. Specifically, the average accuracy is higher than $S_4 \rightarrow T_4$ and $S_5 \rightarrow T_5$ by 0.97% and 2.18% when the size of fine-tuning subset rises to 800 as well. It proves that employing the proposed JATL framework, the useful fault information of the transferred machine is preserved even after extensive fine-tuning. The proposed approach improves diagnosis generalization performance and is better than traditional randomly weight initialization.

These two Cases have fully identified the effectiveness and transferability of the proposed MJDCNN and JATL framework at different levels. The investigations demonstrate that the proposed approach has outstanding fault diagnosis ability in the source domain and has reliable transferability in different target domains.

5.4. The effect of different multilinear features

The MJDCNN for these two diagnosis tasks is made up of four convolutional layers. The effect of the multilinear features from different layers is investigated as follows. As shown in Table 7 and Fig. 14, JD_{14} , JD_{24} , and JD_{34} represent the multilinear features of the last convolutional layer and the first, second, and third convolutional layer, respectively. It can be observed that because features obtained from the first convolutional layer still are much noisy and less discriminative, their prior information is less reliable in the joint adaptive transfer learning part. On the contrary, features extracted from the third convolutional layer are too specific and discriminative to transfer to the diagnosis tasks under other domains. When the domain changes rapidly, the efficiency of JD_{34} decreases quickly.

To further explore the internal mechanism of the joint adaptive transfer learning framework, the intermediate activations of JD_{24} are visualized in Fig. 15. It displays how continuous convolutional layers transform the inputs and helps understand the meaning of each filter of the deep convolutional neural network. If some regions are activated in yellow, some information that facilitates diagnosis is detected at this kernel receptive field. A randomly chosen sample from fault IF under 0 HP is set as an activation input. What's more, models pre-trained on S_3 with and without fine-tuning on the target domain are chosen to make a comparison.

As shown in part bounded by the red line in Fig. 15, the first layer is a collection of various wave peak detectors. At this stage, activation retains almost all the statistical information in the original signal. It is pretty difficult to discriminate features in the first two layers between these two models. Besides, it indicates that even after fine-tuning in another domain, the general bottom fea-

Table 7

Classification accuracy using different multilinear features.

Multilinear features	$S_1 \rightarrow T_0$	$S_2 \rightarrow T_0$	$S_3 \rightarrow T_0$
JD_{14}	96.08%	96.50%	94.85%
JD_{24}	97.54%	97.58%	95.96%
JD_{34}	97.35%	95.27%	95.27%

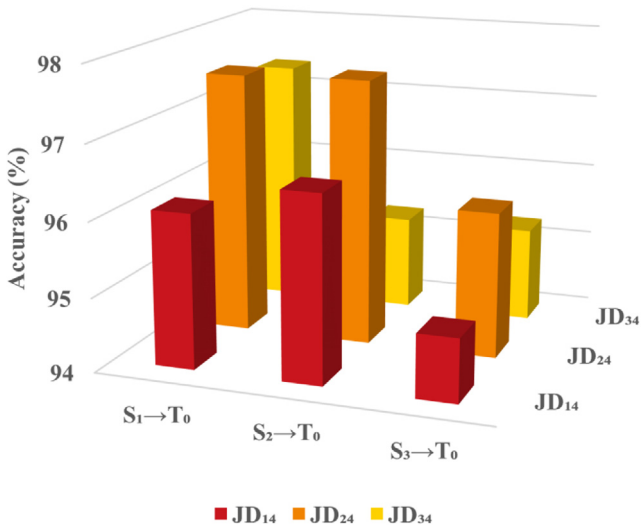


Fig. 14. Comparison results on different multilinear features.

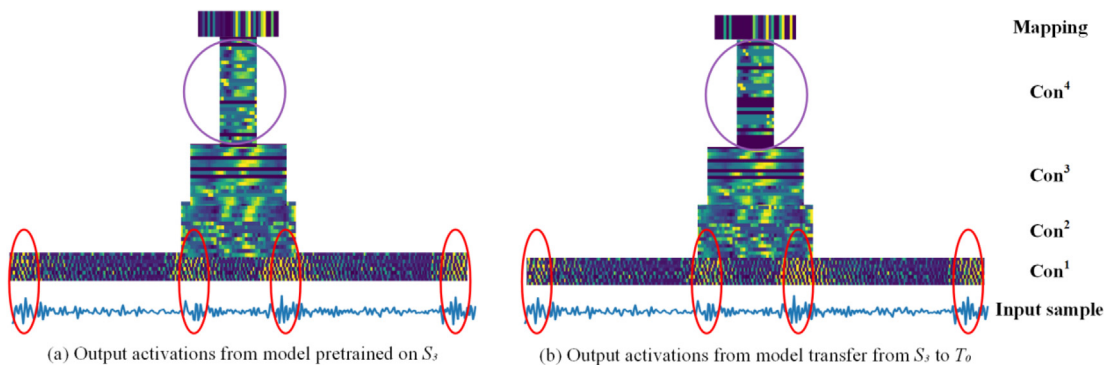


Fig. 15. Comparison visualization results on activations.

tures are similar. Besides, as shown in part bounded by the purple line in Fig. 15, the difference of yellow activation sparsity between these two models increases as layers go deeper. It indicates that higher features are more relevant to specific tasks when adaptively tuned on the target domain. However, with regard to the insufficiency of data in the target domain, extracted specific features are usually no longer reliable. To cope with it, general features obtained by bottom layers are exploited to assist the learning process of higher layers without being distorted by intermedia layers in the proposed approach. Overall, the joint adaptive transfer learning framework aims to make a tradeoff between general features and specific demands of target tasks. In the face of fewer target domain data, the MJDCNN under the proposed framework achieves better transferability and learns discriminative features.

6. Conclusion

This paper proposes a novel end-to-end joint adaptive transfer learning framework for cross-domain machine intelligent diagnosis based on multi-layer feature fusion. An efficient feature fusion between different layers is made by a devised MJDCNN using the multilinear map. It makes utmost utilization of hierarchical information, which embeds the joint distribution of multiple layers into the reproducing kernel Hilbert space without losing early statistical features. Multilinear features significantly improve the diagnosis quality, achieving 99.95% diagnosis accuracy on a multiclassification task. What's more, we build a joint adaptive transfer learning framework aiming at the difficulties such as changes in environmental and operational conditions. It mainly has two stages: first, the base MJDCNN is pre-trained on the source domain. With a joint adaptive fine-tuning strategy, a model initialized with pre-trained weights is transferred to a related yet different task. This framework both considers the domain adaptation needing for the entire model and the different transferability of layers. Experiments illustrate that fusing with cross-domain-invariant features can break the bottleneck of traditional models. As a result, this approach is promising to help apply IFD in practical practice.

Further work is being conducted, including extending the proposed transfer learning framework by further alleviating the need for labeled data in the target domain. Extending the proposed supervised methodology toward unsupervised settings can improve the availability in the practical industry.

CRedit authorship contribution statement

Yimin Jiang: Conceptualization, Methodology, Writing – original draft. **Tangbin Xia:** Methodology, Writing – original draft, Software. **Dong Wang:** Data curation, Supervision, Investigation. **Kaigan Zhang:** Software, Validation, Visualization. **Lifeng Xi:** Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported by National Natural Science Foundation of China (51875359), Natural Science Foundation of Shanghai (20ZR1428600), Shanghai Science & Technology Innovation Center for System Engineering of Commercial Aircraft (FASE-2021-M7),

Ministry of Education-China Mobile Joint Foundation Construction Project (MCM20180703).

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